

Re-write each expression into vertex form by completing the square.

$$\begin{aligned} 1. \quad & x^2 + 14x - 38 \\ & [x^2 + 14x + 49] - 38 - 49 \\ & (x + 7)(x + 7) - 87 \\ & (x + 7)^2 - 87 \end{aligned}$$

$$\begin{aligned} 2. \quad & n^2 - 2n - 3 \\ & [n^2 - 2n + 1] - 3 - 1 \\ & (n - 1)^2 - 4 \end{aligned}$$

$$\begin{aligned} 3. \quad & f^2 + 14f - 51 \\ & [f^2 + 14f + 49] - 51 - 49 \\ & (f + 7)^2 - 100 \end{aligned}$$

$$\begin{aligned} 4. \quad & a^2 + 2a + 20 \\ & [a^2 + 2a + 1] + 20 - 1 \\ & (a + 1)^2 + 19 \end{aligned}$$

$$\begin{aligned} 5. \quad & b^2 - 4b + 4 \\ & b^2 - 4b + 4 \\ & (b - 2)^2 \end{aligned}$$

$$\begin{aligned} 6. \quad & c^2 + 6c + 8 \\ & [c^2 + 6c + 9] + 8 - 9 \\ & (c + 3)^2 - 1 \end{aligned}$$

$$7. \quad \begin{array}{l} g^2 + 4g + 4 \\ (g + 2)^2 \end{array}$$

$$8. \quad \begin{array}{l} t^2 - 10t \\ [t^2 - 10t + 25] - 25 \\ (t - 5)^2 - 25 \end{array}$$

$$9. \quad \begin{array}{l} z^2 + 4z - 6 \\ [z^2 + 4z + 4] - 6 - 4 \\ (z + 2)^2 - 10 \end{array}$$

$$10. \quad \begin{array}{l} d^2 + 6d + 8 \\ [d^2 + 6d + 9] + 8 - 9 \\ (d + 3)^2 - 1 \end{array}$$